

Fucheng Guo

Raleigh, NC · fguo22@ncsu.edu · [Personal Website](#) · [Google Scholar](#)

EDUCATION

North Carolina State University

Ph.D. in Computer Science (GPA 4.0/4.0)

Raleigh, NC

08/2023 – present

- Advised by Frank Mueller and Yuan Liu.

Xi'an Jiaotong University

Ph.D. in Power Engineering and Engineering Thermophysics

Xi'an, China

2017 – 2023

Xi'an Jiaotong University

B.S. in Energy and Power Engineering

Xi'an, China

2013 – 2017

RESEARCH EXPERIENCE

North Carolina State University

08/2023 – present

Quantum versus classical algorithms for lossy GBS

09/2026 – present

- Studied the quantum–classical performance gap in Gaussian boson sampling under constant photon loss by comparing tensor-network classical simulation algorithms with quantum sampling in a previously underexplored regime.

Memory-optimal sequential Gaussian synthesis

01/2026 – 08/2026

- Proposed a sequential transmission protocol for inter-module communication in modular quantum systems.
- Developed methods to determine the optimal memory cost and near-optimal emission orders, together with a general algorithm for synthesizing sequential Gaussian circuits.

Concatenated dual displacement code for CV QEC

05/2025 – 12/2025

- Proposed a concatenated QEC scheme that circumvents the Gaussian no-go limitation, with an outer code correcting large displacement errors beyond the correctable range of an inner GKP code.
- Evaluated its error-suppression performance under Gaussian displacement noise using Monte Carlo simulation.

Performance analysis of HPC programs (advised by Xu Liu)

08/2023 – 08/2024

- Analyzed data-movement patterns in HPC programs using binary instrumentation tools and investigated source-level optimizations to reduce redundant data transfers.

Xi'an Jiaotong University

2019 – 2023

Numerical simulation of thermal and mechanical systems

2019 – 2023

- Performed numerical simulations of complex mechanical structures, heat-transfer processes, and phase-change phenomena.

SKILLS

- **Quantum information:** continuous-variable quantum computing, quantum error correction, Gaussian boson sampling, state preparation, circuit synthesis
- **Methods:** symplectic methods, Monte Carlo simulation, tensor-network simulation, numerical optimization
- **Software:** Python (NumPy, SciPy), QuTiP, Qiskit, Strawberry Fields, Linux/HPC

PUBLICATIONS

- **Fucheng Guo**, Frank Mueller, and Yuan Liu, “Concatenated dual displacement code for continuous-variable quantum error correction,” *Physical Review Research* **8**, 023158 (2026).
- **Fucheng Guo**, Frank Mueller, and Yuan Liu, “Memory-Optimal Sequential Synthesis of Multimode Gaussian Transformations,” arXiv preprint, 2026.

PROFESSIONAL SERVICE

- Reviewer for Physical Review Letters, PRX Quantum, Physical Review A, Physical Review Applied, Physica Scripta, Quantum Science and Technology, IEEE Transactions on Radar Systems, and HPCA.

HONORS AND AWARDS

- China National Scholarship 2022
- National Second Prize, China Undergraduate Mathematical Contest in Modeling (CUMCM) 2014